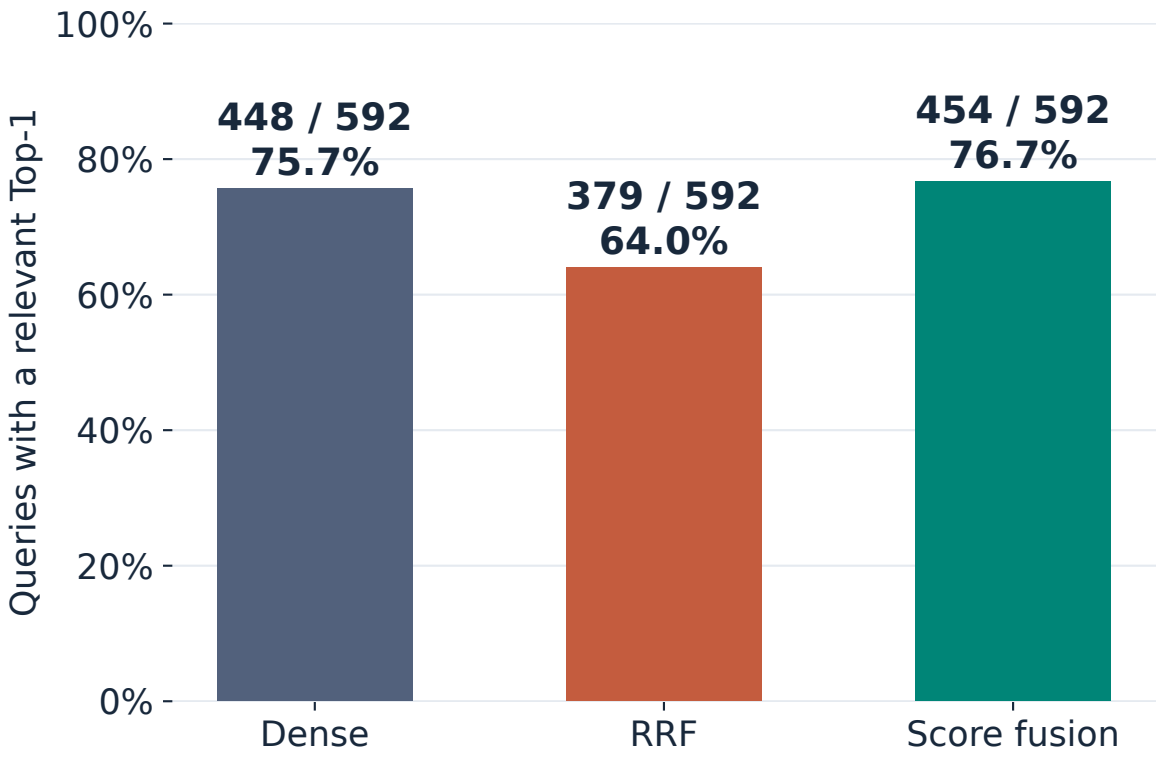
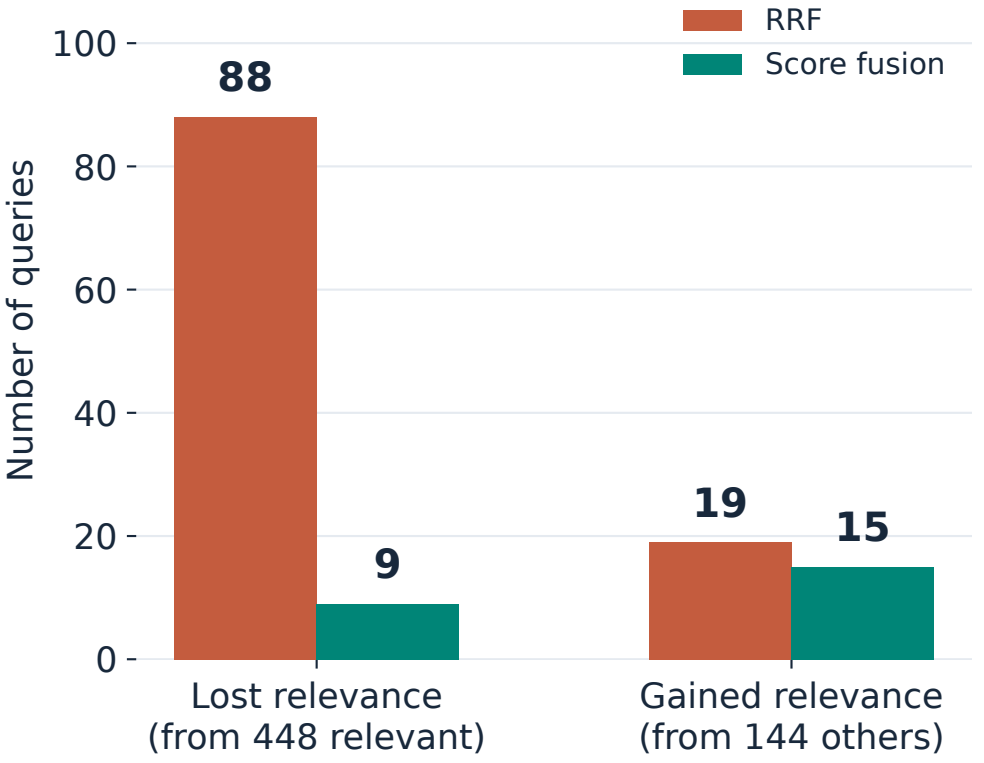
Score fusion preserves relevant first results

Ko-StrategyQA dev | 592 queries | BGE-M3 | independently ranked methods

Relevant Top-1



Changes relative to Dense



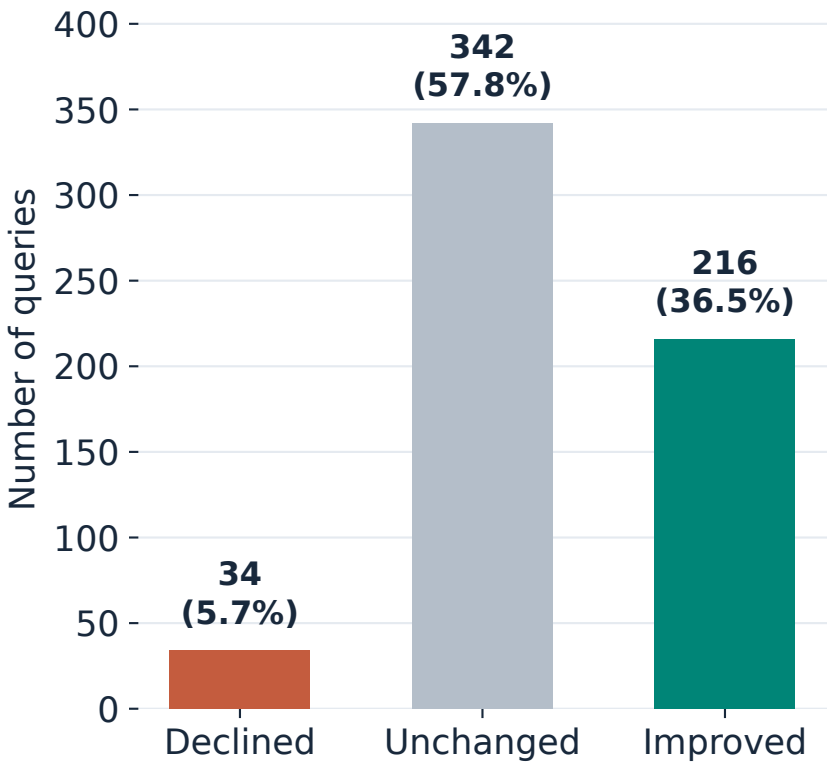
79 fewer losses − 4 fewer gains
= 75 more relevant first results

Relevant = positive qrel. Unjudged documents are not counted as relevant; they are not proven irrelevant.

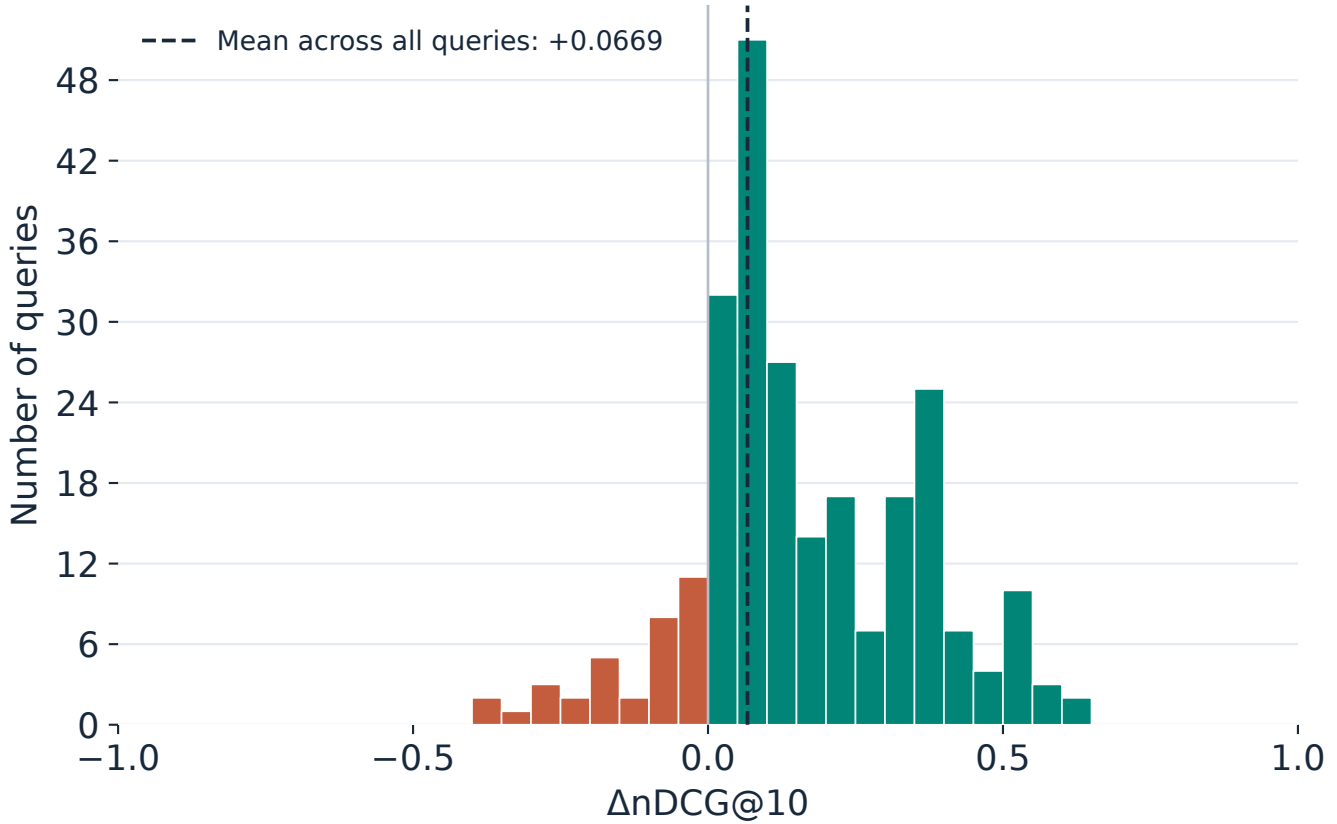
More queries improve than decline

Per-query ΔnDCG@10 = Score fusion − RRF | Ko-StrategyQA dev | 592 queries

All queries



250 changed queries

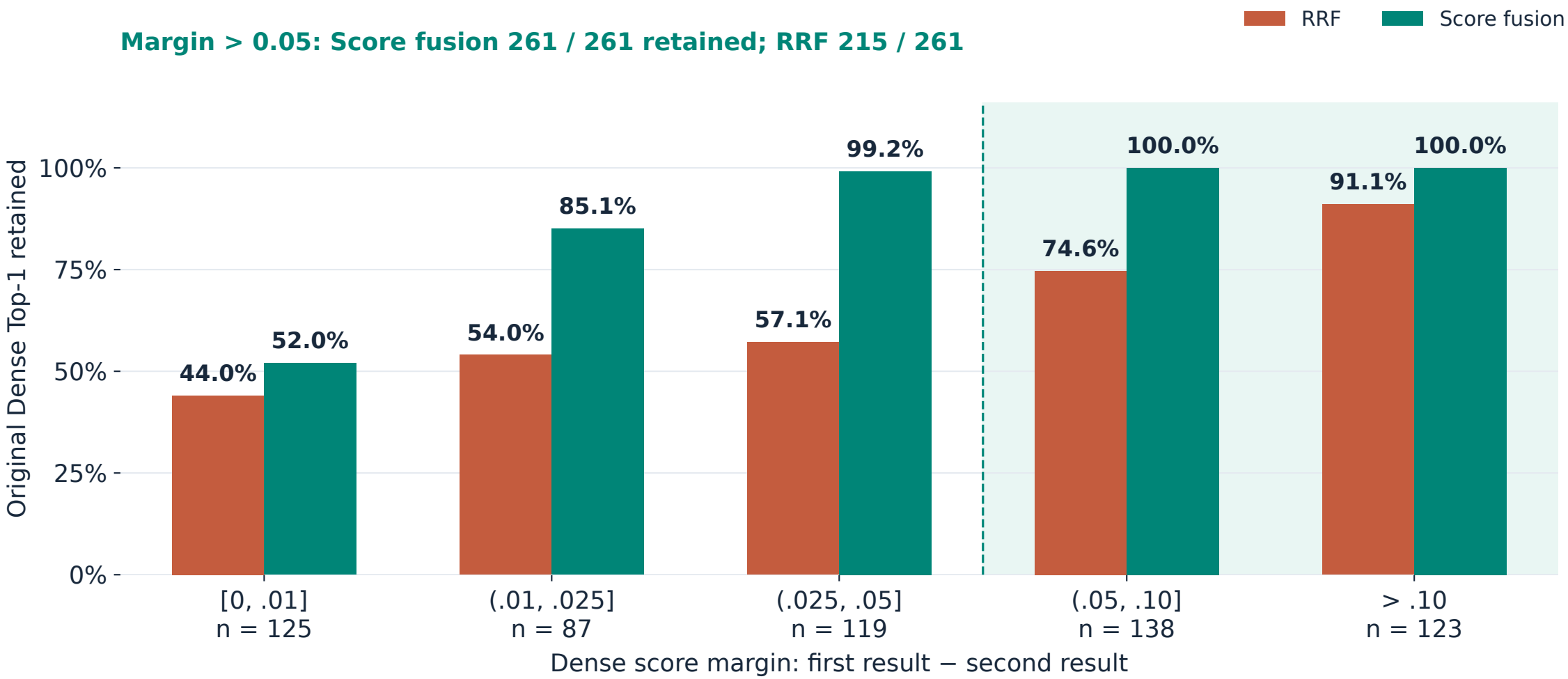


The 342 unchanged queries are excluded from this histogram.

Nonzero bins have width 0.05. These are descriptive query-level effects, not independent samples.

Large Dense margins protect the first result

Same-document Top-1 retention | not Top-1 relevance | Ko-StrategyQA dev



BM25 adds at most 0.05, so it cannot reverse a larger Dense score gap.

Observed rates; no fitted trend. Preserving the original first result does not guarantee relevance.